EDU360: Virtual K-12 Academy — Role-Wise Daily Activities & Workflows

Knowledge Transfer (KT) Handbook • Black & White Executive Edition • The 8 Canonical Roles • Daily Routines & Closed Loops

> **Executive Overview:** This document is formatted strictly in **clean, high-contrast Black & White**. It breaks down the entire EDU360 virtual school system **role by role**. For each of the 8 canonical roles, it outlines their **Daily Morning-to-Evening Schedule**, their **Closed-Loop Workflows**, and **exact KT speaking points** in simple English.

Role 1: Super Admin / Principal (Portal: /admin)

SUPER ADMIN / PRINCIPAL

Core Mission: Provides overall school oversight, monitors real-time operational safety, handles legal document seals, and manages executive task escalations.

Daily Morning-to-Evening Activity Routine

- **08:30 AM — Priority Triage Inspection:** Checks the Red/Orange/Green command cards. Evaluates critical flags (e.g., student absent 3+ days or sudden grade drops ≥ 15%).
- **10:30 AM — Admin Tasks & Escalation Cockpit:** Reviews escalated tasks from coordinators. Approves emergency budget exceptions or faculty coverage.
- **01:30 PM — Official Document Vault Seals:** Digitally signs and officially seals student Transfer Certificates (TC), Bonafide certificates, and admission intake records.
- **04:30 PM — Statutory CBSE Audit Log Review:** Performs 1-click export of system audit logs with cryptographic checksums for board compliance.

Key Closed-Loop Workflows Owned by Super Admin

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Workflow A: Priority Triage Auto-Escalation (Absenteeism & Grade Drops)
Step 1: System background rule runs every 60 seconds.
Step 2: Detects student with 3 consecutive unexcused online absences.
Step 3: Automatically creates RED ALERT on Admin dashboard + Priority WhatsApp to parent.
Step 4: ✓ Welfare Task Triggered: Assigns mandatory 8-hour welfare phone check to Class Teacher.
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Workflow B: Dual-Signature Student Exit & Transfer Certificate (TC) Vault
Step 1: Clearance Audit: Accounts verifies zero arrears, Academic verifies all term grades recorded.
Step 2: Parent reviews digital TC in Parent App and submits Digital OTP Sign-off (timestamped in SIS).
Step 3: Physical hardcopy is handed over / couriered to parent, and parent signs acknowledgment slip.
Step 4: ✓ Permanent Legal Seal: Admin scans signed slip into the SIS Vault. Permanent dual-signature lock
prevents future legal disputes.
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> 🎙 **What to Say in KT:** "The Super Admin has a bird's-eye view. They do not do manual data entry. Instead, the Priority Triage engine flags problems automatically (like 3 absences in a row), and they hold the legal master key for official document seals."

Role 2: Academic Coordinator (Portal: /academic_coordinator)

ACADEMIC COORDINATOR /academic_coordinator

Core Mission: Ensures teaching quality, monitors syllabus pacing against academic calendar weeks, arranges emergency substitute teachers, and protects teachers from burnout.

Daily Morning-to-Evening Activity Routine

- **08:15 AM — Morning Teacher Leave & Substitution Review:** Scans sick leave requests filed by faculty. Uses AI Timetable Governor to assign qualified peer substitutes.
- **11:00 AM — Curriculum Pacing Surveillance:** Reviews weekly progress bars comparing scheduled units vs units actually delivered by teachers.
- **02:30 PM — Faculty Task Relief Queue:** Reviews SLA pause requests submitted by teachers. Grants extensions (+12h to +72h) or migrates tasks to peer faculty.
- **05:00 PM — Remedial Focus Table Audit:** Checks Class 8-B Mathematics and other sections to ensure Day 1–7 remedial drills are active.

Key Closed-Loop Workflows Owned by Academic Coordinator

Workflow A: AI Timetable Governor & Dynamic Substitution Cover
Step 1: Teacher submits urgent sick leave in teacher portal.
Step 2: AI Timetable Governor scans master schedule for teachers who are free during those periods and qualified in compatible subjects.
Step 3: Coordinator clicks [Assign Cover] on the ranked substitute list.
Step 4: ✓ **Instant Schedule Sync:** Substitute teacher portal receives purple SUBSTITUTION_COVER badge, and student class links update immediately.

Workflow B: Human-in-the-Loop Faculty SLA Relief State Machine
Step 1: Teacher assigned a 48h grading task experiences an emergency or workload strain.
Step 2: Teacher clicks [Request Task Relief] → SLA timer stops immediately with ZERO penalty.
Step 3: Task status changes to relief_requested and appears in Coordinator's relief queue.
Step 4: ✓ **Coordinator Decision:** Coordinator grants +12h, +24h, +48h, or +72h extension, or migrates the task to a peer teacher without penalizing the original teacher.

> 🎙 **What to Say in KT:** "Show how the Academic Coordinator manages school operations without stress. Morning substitutions take 1 click instead of 20 phone calls. And if teachers have an emergency, the coordinator can pause their deadline clock and give extra time."

Role 3: Exam Coordinator (Portal: /exam_coordinator)

EXAM COORDINATOR

Core Mission: Controls examination schedules, locks question paper blueprint schemas, enforces accounts-gated hall tickets, and manages paper grading distribution.

Daily Morning-to-Evening Activity Routine

- **09:00 AM — Blueprint Lock Verification:** Confirms question paper schema locks (Recall, Application, Analysis points) 72 hours before term exams.
- **11:30 AM — Accounts-Gated Hall Ticket Audit:** Verifies student clearance against the Accounts ledger. Manages cleared lists for exam portal access.
- **02:00 PM — Virtual Invigilation Rostering:** Assigns faculty proctors to virtual examination breakout rooms.
- **04:30 PM — Paper Distribution Roster:** Routes uploaded student exam answer scripts to designated subject teachers with a 48-hour grading SLA.

Key Closed-Loop Workflows Owned by Exam Coordinator

Workflow A: Accounts-Gated Hall Ticket Locking
Step 1: Student attempts to download Term Exam Hall Ticket in student portal.
Step 2: Exam engine queries Accounts ledger in real-time.
Step 3: If unpaid dues exist and no waiver is granted: Hall Ticket download is locked.
Step 4: ✓ **Real-Time Clearance:** When Accounts Coordinator logs counter payment or toggles exam_clearance = True, Hall Ticket unlocks immediately!

Workflow B: 72-Hour Blueprint Freeze & Moderation Distribution
Step 1: Exam blueprints uploaded with marks allocation per question.
Step 2: 72 hours before exam, blueprints are permanently locked against unauthorized changes.
Step 3: After exam, student answer scripts are batched and routed to teachers with an automated 48-hour SpeedGrader countdown.

> 🎙 **What to Say in KT:** "The Exam Coordinator ensures exam integrity. Question blueprints freeze 72 hours early so nobody can leak or tamper with papers. And hall tickets are tied directly to the fee ledger—if a student clears dues, access unlocks instantly."

Role 4: Accounts Coordinator (Portal: /accounts_coordinator)

ACCOUNTS COORDINATOR (BURSAR)

Core Mission: Manages tuition fee billing, tracks overdue aging ledgers, records counter payments with UTR reconciliation, and approves fee hardship waivers.

Daily Morning-to-Evening Activity Routine

- **08:45 AM — Daily Arrears Aging Check:** Scans overdue fee buckets: 0–15 Days, 16–30 Days, and 30+ Days Overdue.
- **11:00 AM — Admissions Fee Intake Review:** Checks new converted student records from Admissions. Verifies paid UPI transactions or pending admission fee invoices.
- **01:30 PM — Counter Collections & UTR Reconciliation:** Enters offline payments (Cash, Cheque, NEFT bank transfers), records UTR reference, and prints receipts.
- **04:00 PM — Management Concession & Hardship Waivers:** Authorizes fee waivers with audit reason codes. Toggles exam clearance overrides for distressed families.

Key Closed-Loop Workflows Owned by Accounts Coordinator

Workflow A: Admissions-to-Accounts Handshake & Ledger Routing
Step 1: Admissions Coordinator converts an enquiry to SIS.
Step 2: If fee was paid: Reconciled PaymentTransaction created with UTR (UPI/timestamp/ADM) + receipt code **REC-ADM-2026-X.**
Step 3: If fee was unpaid: FeeDefaulter created with is_admission_fee = True and enters Arrears Aging Ledger.
Step 4: **✓ Bursar Control:** Bursar collects payment at counter, grants a concession, or unlocks exam access with one click.

Workflow B: Institutional Exam Clearance Override
Step 1: Student owes fee balance but needs to sit for exams.
Step 2: Bursar opens fee defaulter row and toggles **[Exam Clearance = Granted].**
Step 3: **✓ Financial Balance Preserved:** Student hall ticket unlocks immediately, while the unpaid balance remains active on the ledger for future collection.

> 🎙 **What to Say in KT:** "Accounts is completely synchronized with Admissions. When a student is enrolled, the bill appears here automatically. The bursar can track overdue payments by age, issue receipts with UTR codes, or allow a student to take exams even if fees are delayed."

Role 5: Admissions Coordinator (Portal: /admissions_coordinator)

ADMISSIONS COORDINATOR

Core Mission: Drives new student enrollment, logs parent telephonic outreach, tracks lead interest temperature, and executes 1-click atomic conversion into active SIS students.

Daily Morning-to-Evening Activity Routine

- **09:00 AM — New Lead Triage:** Inspects overnight website enquiries. Prioritizes leads based on interest temperature (Hot, Warm, Cold).
- **11:30 AM — Outreach Calling & Notes Logging:** Calls prospective parents. Records call transcripts, answers curriculum questions, and sets next-touch dates.
- **02:30 PM — Virtual Academy Tours & Entrance Tests:** Schedules virtual campus tour webinars and monitors online entrance assessments.
- **04:30 PM — 1-Click SIS Conversion & Intake Promotion:** Converts qualified applicants into active SIS students with uploaded intake documents.

Key Closed-Loop Workflows Owned by Admissions Coordinator

Workflow A: 1-Click Atomic SIS Student Conversion
Step 1: Student clears entrance evaluation (Stage: Entrance Cleared).
Step 2: Coordinator selects section (e.g. Class 8-B), enters admission fee amount, and clicks **[Convert to SIS].**
Step 3: System creates permanent Student record with unique Roll No and Admission ID (EDU-2026-XXX).
Step 4: Intake documents (Birth Cert, Previous Marks) promoted to OfficialDocument in SIS Vault (Pending Admin Seal).
Step 5: **✓ Cross-Portal Handshake:** Admission fee invoice sent straight to Accounts Coordinator's ledger!

Workflow B: Statutory Lead Drop Auditing
Step 1: Parent unresponsive after 3 attempts or declines admission.
Step 2: Coordinator can only mark lead dropped by selecting a mandatory statutory reason (fees_high, distance, curriculum_fit).
Step 3: **✓ Reversible:** Dropped leads can be restored at any time if parent calls back.

> 🎙 **What to Say in KT:** "The Admissions CRM is not a disconnected sales tool. Converting an enquiry instantly creates the student's real profile, vaults their certificates, and sends their bill to Accounts with zero manual re-typing."

Role 6: Teacher Portal (Portal: /teacher)

TEACHER PORTAL (MRS. ANANYA VERMA)

Core Mission: Delivers live video classes, monitors automatic join attendance, evaluates homework with an in-browser digital red pen, and guides failing students through 1-week remedial sprints.

Daily Morning-to-Evening Activity Routine

- **08:30 AM — Timetable & Cover Inspection:** Checks daily class periods and any assigned `SUBSTITUTION_COVER` slots.
- **09:00 AM — Live Video Class & 10-Min Auto-Attendance:** Launches class. System records join timestamps automatically.
- **11:30 AM — SpeedGrader Canvas Evaluation:** Evaluates uploaded worksheets inside browser with digital red pen, checkmarks,

and deduction stamps.

- **02:00 PM — 1-Week Remedial Focus Table Review:** Inspects Class 8-B Math student rough notebook photos against rubric checkpoints. Approves satisfactory work.
- **04:00 PM — 1-Click Syllabus Delivery Logging:** Taps completed curriculum units to update the Academic Coordinator's pacing surveillance chart.

Key Closed-Loop Workflows Owned by Teacher

Workflow A: 10-Minute Online Class Auto-Attendance Rule
Step 1: Teacher starts live video session (T = 0 mins). Webhook listener activated.
Step 2: Student clicks Join: Joined ≤ 10m = **PRESENT;** Joined 10–25m = **LATE;** Joined > 25m or no join = **ABSENT.**
Step 3: ✓ **Zero Class Time Wasted:** Saves 15 minutes of roll call. 3 consecutive absences trigger Principal RED alert.

Workflow B: The 1-Week Remedial Focus Sprint (≤ 70% Score)
Step 1: Students scoring ≤ 70% in Mathematics exams auto-enroll in 7-day sprint (Students > 70% are Exempt).
Step 2: Student solves daily micro-problem in **physical rough notebook** and uploads photo.
Step 3: Teacher opens inspection modal: views assigned problem and rubric checkpoint alongside student handwritten steps.
Step 4: ✓ **Parent Nightly Sign-off:** Teacher approves work → Parent verifies physical notebook at home for 10 minutes and signs off in app → Day 2 unlocked!

> 🎙 **What to Say in KT:** "The Teacher portal is designed to eliminate clerical work. Roll call is automatic. Grading is done right in the browser with a digital red pen. And weak students get a structured 7-day plan with physical notebook practice verified by parents."

Role 7: Student Portals (Secondary: /student • Junior K–5: /junior)

STUDENT PORTALS (SECONDARY & JUNIOR K–5)

Core Mission: Attends live video classes, uploads homework worksheets, practices remedial drills in a physical rough notebook, and uses the AI Socratic Tutor for conceptual hints.

Daily Morning-to-Evening Activity Routine

- **08:45 AM — Daily Timetable & 1-Click Live Join:** Checks today's periods and clicks [Join Video Class] to be marked Present.
- **01:00 PM — Homework Worksheet Submission:** Uploads handwritten homework photos before the strict 5:00 PM cutoff.
- **03:30 PM — AI Socratic Tutor Doubt Solving:** Asks math doubts to AI tutor; receives guiding hints and conceptual questions without direct answers.
- **05:30 PM — 1-Week Focus Sprint Rough Notebook Practice:** Solves Day N micro-problem in physical rough notebook and uploads photo.
- **Junior K–5 Students:** Explore adventure map timetable, open Magic Booster formula chests, and complete daily academic quests.

Key Closed-Loop Workflows Owned by Student

Workflow A: 15-Minute Missed Homework Cutoff & Backlog Evidence
Step 1: Daily homework has a strict cutoff (e.g. 5:00 PM).
Step 2: Background worker sweeps unsubmitted homework every 15 minutes.
Step 3: ✓ **State Transition:** At 5:01 PM, unsubmitted tasks transition from PENDING to BACKLOG with evidence ID **#EVD-BKLOG-X.**
Step 4: Appears in Parent App as unresolved backlog and in teacher's meeting dossier.

> 🎙 **What to Say in KT:** "For older students, everything is clean and fast—1-click video join, homework dropzone, and an AI tutor that teaches instead of giving answers. For younger kids in K–5, school feels like an exciting adventure quest."

Role 8: Parent Portal (Portal: /parent)

PARENT PORTAL

Core Mission: Co-tracks student academic recovery through nightly rough-book inspection, monitors live class attendance, and pays fee invoices instantly with zero notification spam.

Daily Morning-to-Evening Activity Routine

- **08:50 AM — Anti-Spam Moving Announcement Slider:** Reads routine circulars gliding on top carousel without notification fatigue.
- **10:15 AM — Live Class Presence Radar:** Checks if child joined live class on time or was flagged late/absent.
- **07:30 PM — 10-Minute Nightly Rough-Book Sign-Off:** Inspects child's physical rough notebook for Day N math remedial drill, reviews teacher notes, and signs off.
- **Monthly — 2-Second Fee Checkout:** Pays fee installment with 1-click gateway and downloads digitally sealed PDF receipt.

Key Closed-Loop Workflows Owned by Parent

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Workflow A: 10-Minute Nightly Rough-Book Co-Tracking Sign-Off
Step 1: Parent opens [10-Minute Nightly Co-Tracking] card at 7:30 PM.
Step 2: Sees child's daily math problem and teacher's approval note.
Step 3: Parent checks child's physical rough notebook for 10 minutes.
Step 4: ✓ Milestone Sign-off: Parent taps [Sign-Off Rough Book] in app ➔ Unlocks Day N+1 practice drill!
```

🎙 **What to Say in KT:** "We respect parents' time. Instead of sending 15 noisy WhatsApp alerts a day, routine notices glide on a top slider. We ask for just 10 minutes at night to check their child's rough notebook and sign off in the app."

Summary Table: Real Code vs Gemini AI vs Fallbacks

System Engine	How it Operates in Real Production	Fallback Mode (If AI / Network is Down)
Priority Triage (Red/ Orange)	**100% Real Code:** FastAPI background rules scanning attendance streaks, grade drops, and fee dues every 60 seconds.	Always active. Zero external dependencies.
10-Min Auto-Attendance	**100% Real Code:** Webhook listener calculating delta between student join time and class start time.	Always active. Zero external dependencies.
SpeedGrader Digital Red Pen	**100% Real Code:** HTML5 Canvas engine with variable red-pen stroke weights, stamps, and rubric deductions.	Always active. Zero external dependencies.
Admin Tasks & Relief Engine	**100% Real Code:** Event-driven task generation with SLA timers, human relief state machine, and escalation paths.	Always active. Runs directly inside database.
Admissions-to-Accounts Handshake	**100% Real Code:** Atomic database transaction creating Student profile, vault documents, and fee ledger entry.	Always active. Zero external dependencies.
1-Week Remedial Sprint	Calls Google Gemini 1.5/2.0 API with weak concepts to generate 7-day micro-practice syllabus.	Uses an algorithmic math curriculum template so the 7-day sprint generates instantly without failure.
Rough Work Pre-Check	Uses Gemini Vision to read student handwritten rough work photos against the teacher's rubric checkpoint.	Shows the question and rubric side-by-side so the teacher can verify the paper manually in seconds.
AI Socratic Doubt Tutor	Calls Gemini Chat API instructed to guide students with step-by-step hints without giving direct answers.	Uses a curated dictionary of Socratic questions sorted by math chapters.

Live Product Demo Cheat-Sheet (5-Minute Screen Share)

Portal & Role	Login Credentials	Exact Screen & Action to Show During Screen Share
1. Super Admin `/admin`	`admin` `Password123`	Show top **Priority Triage Cards** (Red alert for 3 absences). Open **Admin Tasks** table ➔ Show SLA countdown timer, [Request Relief], and 1-click statutory audit export.
2. Admissions `/admissions_coordinator`	`admissions_coordinator` `Password123`	Open enquiry pipeline. Click on a student lead ➔ Click **[Convert to SIS]**. Show that Roll No is minted, documents promoted to Vault, and fee invoice sent to Accounts!
3. Accounts `/accounts_coordinator`	`accounts_coordinator` `Password123`	Show the **Arrears Aging Ledger**. Find the converted student's admission fee. Demonstrate collecting counter payment with UTR or toggling [Exam Clearance Override].
4. Teacher `/teacher`	`teacher_mat` `Password123`	Filter Focus Table by `Class 8-B`. Show ≤ 70% remedial enrollment. Click Day 2 rough work modal. Open SpeedGrader ➔ Draw red-pen checkmark on canvas & click [Finalize Grade].
5. Parent `/parent`	`parent` `Password123`	Show the top moving announcement carousel. Open the **10-Minute Nightly Co-Tracking Card** and click [Sign-off Rough Book]. Show live video class radar.
6. Junior K-5 `/junior`	`junior` `Password123`	Show the illustrated **Adventure Quest Timetable**, magic booster chests, and visual live class launchpad designed for elementary children.

Universal Login Password: Every single test account uses the universal password: `Password123`.
Offline Login Resilience: Even if the backend server is stopped, the frontend uses an instant 3-second offline cache so you will never get locked

> out during a demo.